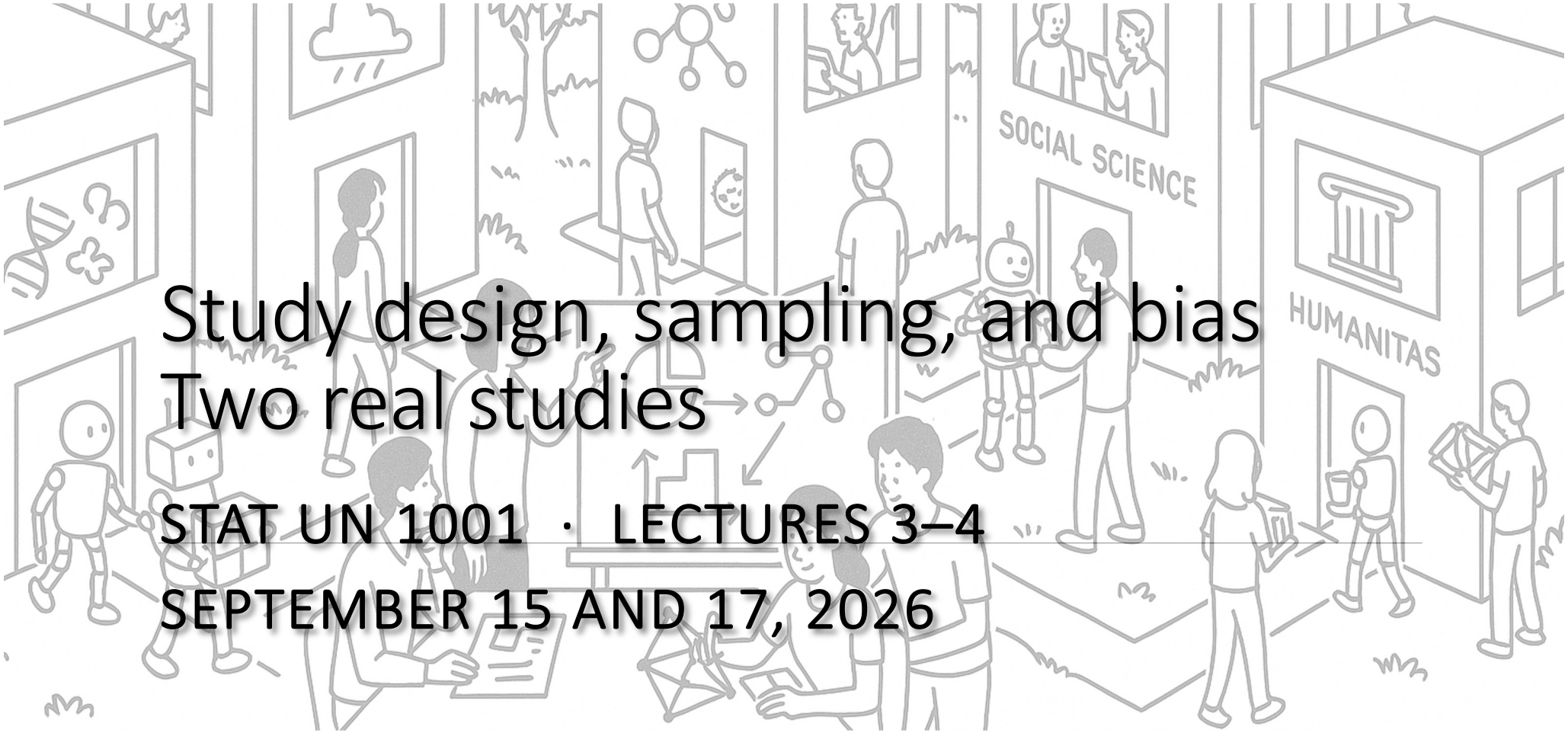


Study design, sampling, and bias
Two real studies

STAT UN 1001 · LECTURES 3–4

SEPTEMBER 15 AND 17, 2026

Last week Important concepts

A statistical question connects a population to a variable we want to understand.

Cases or individuals	The people, objects, places, or time periods represented by rows of data
Variables and values	A variable names what varies. A value records what we observed for one case.
Sample and population	The sample provides the data. The population defines the group we hope to describe.

A credible claim must match the evidence and how the data were produced.

Sample vocabulary in one example

A random sample of 1,155 Americans reported their hours of relaxation after an average workday. The sample mean was 1.65 hours.

Observation	One sampled American
Variable	Hours available to relax
Statistic	The sample mean, 1.65 hours
Parameter	The unknown mean for all Americans in the target population

A random sample of 1,155 Americans reported how many hours they had to relax after an average workday. The sample mean was 1.65 hours. Which classification is correct?

A sampled American is a statistic; hours of relaxation is an observation; 1.65 is a parameter; the population mean is a variable.	4 respondents	6 %	
A sampled American is a variable; hours of relaxation is a parameter; 1.65 is an observation; the population mean is a statistic.	11 respondents	16 %	
A sampled American is an observation; hours of relaxation is a variable; 1.65 is a sample statistic; the mean for all Americans is a population parameter.	49 respondents	**72 %**	
A sampled American is a parameter; hours of relaxation is a statistic; 1.65 is a variable; the population mean is an observation.	4 respondents	6 %	

Topics from the flipped activity

Cases, variables, values, data matrices, and variable types

Observational studies versus experiments

Populations, samples, observations, statistics, and parameters

Association, causation, and confounding

Sampling methods and sources of sampling bias

Random assignment, controls, placebo, blinding, and blocking

Let’s take a quick look at the flipped quiz.

TA Office Hours

Office Hours

The following office hours are meant for you to ask our TA questions about the flipped learning activities. Please be sure to bring your work and share your attempts first.

Tuesday (In-person) - Room 320 Uris inside the 324 Suite (it is **to the left** when you exit the elevator; the first door **on the right side** of the corridor)

~~8-9 Zara~~
1-2 Gina
2-3 Wenjin

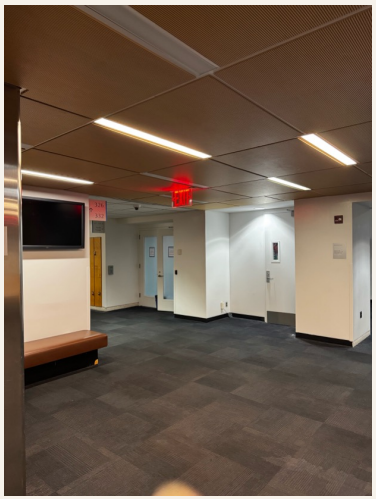
Wednesday (In-person) - Room 324 Uris Open Area (it is **to the left** when you exit the elevator; the first door **on the right side** of the corridor)

10-11 Zara

Friday (Zoom)
10-11 Wenjin
12-1 Gina
1-2 Zara

Zoom Information

TA Office Hours







Cases, variables, and values in a data matrix

Hypothetical data from a statistics practice-tool study

Student	Section	Prior quiz	Practice tool	Chapter 1 score
A	001	14	Yes	18
B	002	11	No	13
C	001	16	Yes	19
D	002	10	No	12

Values

Cases

Students A–D

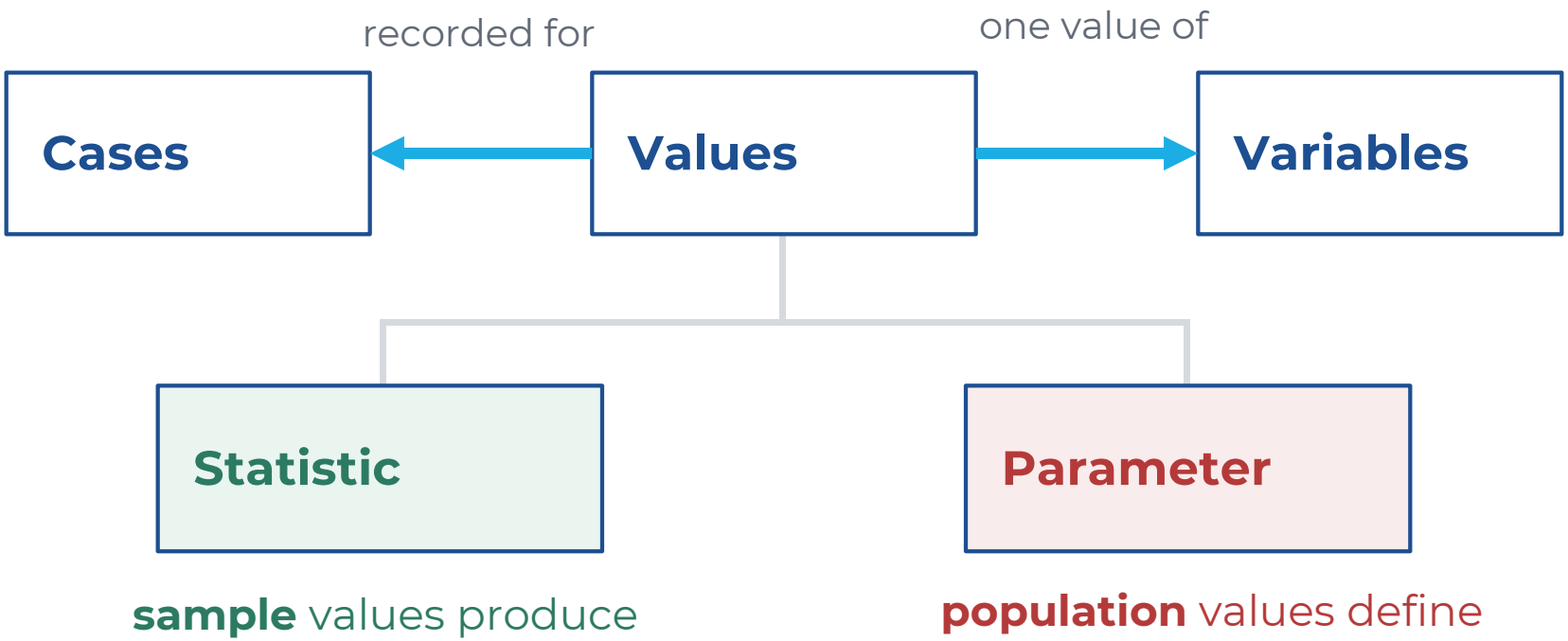
Variables

The column headings

One value

Student B scored 13

Concept map Cases to conclusions



A statistic summarizes observed sample values. A parameter describes the target population.

Statistics and parameters

Research question

What is the average Chapter 1 score among all STAT UN 1001 students?

Statistic

Mean score among the students in the sample

Computed from observed data

Parameter

Mean score among every student in the target population

Usually unknown

Variable types

Categorical

Values place cases into groups or categories

Nominal

No natural order
Example: lecture section

Ordinal

Ordered categories
Example: confidence rating

Numerical

Values record meaningful quantities

Discrete

Countable values
Example: problems completed

Continuous

Measured along a scale
Example: study time

Variable classification Round 1

For each variable, choose categorical or numerical. Then give the subtype.

Lecture section

Number of practice problems completed

Minutes spent studying

Confidence: low, medium, or high

Chapter 1 quiz score from 0 to 20

Used the practice tool: yes or no

Variable classification Tricky examples

Student ID: 384921

Categorical label. Averaging IDs has no meaning.

Completion rate: 80%

Numerical. Differences and ratios can carry meaning when the denominator is clear.

Satisfaction rating: 1 to 5

Usually categorical ordinal. The spacing between categories may not be equal.

Class year: 1, 2, 3, 4

Categorical ordinal when the numbers label first-year through fourth-year students.

The same symbols can represent different variable types. Context determines the meaning.

Observational studies and experiments

Urban rat records

Researchers analyzed complaints and official sightings. They did not assign cities to temperature conditions.

Observational study

The data can show association. Other explanations may remain.

Vitamin C trial

Researchers assigned volunteers to placebo or vitamin-C treatments and measured cold outcomes.

Randomized experiment

Random assignment helps separate the treatment effect from confounding variables.

What are the variables and cases in these studies? How would you organize the data?

Cases	Variable 1				

Cases	Variable 1				

Random sampling and random assignment

RANDOM SAMPLING

Who enters the study?

A probability sample can support generalization from the sample to the population.

Main threat: selection bias

RANDOM ASSIGNMENT

Who receives each treatment?

Random assignment can support a causal comparison among the study participants.

Main threat: confounding

The chia-seed study randomly assigned volunteers. It strengthened causality but did not make the volunteers representative of all adults.

Probability sampling designs Individual selection

Simple random sample

Use the full course roster. Select 60 students so every set of 60 has the same chance of selection.

Strength Known and equal inclusion probabilities

Systematic sample

Choose a random starting point on the roster, then select every kth student.

Watch for A roster pattern that matches the sampling interval

Both designs select individual students directly from a complete sampling frame.

Stratified, cluster, and multistage sampling

Stratified sample

Separate students by lecture section or prior performance, then randomly sample within every group.

Purpose: representation and precision

Cluster sample

Randomly select whole recitation sections, then invite every student in the selected sections.

Purpose: lower collection cost

Multistage sample

Randomly select recitation sections, then randomly select students within those sections.

Purpose: balance cost and coverage

Strata all contribute observations. A cluster sample observes only the selected clusters.